

Practical 4

4. Capture and analyse network packets using Wireshark. Analyse the packets captured.

- Wireshark is a powerful, open-source network protocol analyser used to capture and inspect packets at a very detailed level. It helps in:
 - ❖ **Troubleshooting network issues**
 - ❖ **Monitoring traffic flow**
 - ❖ **Analysing protocols**
 - ❖ **Identifying suspicious or malicious communication**
 - ❖ **Learning how networking works at the packet level**

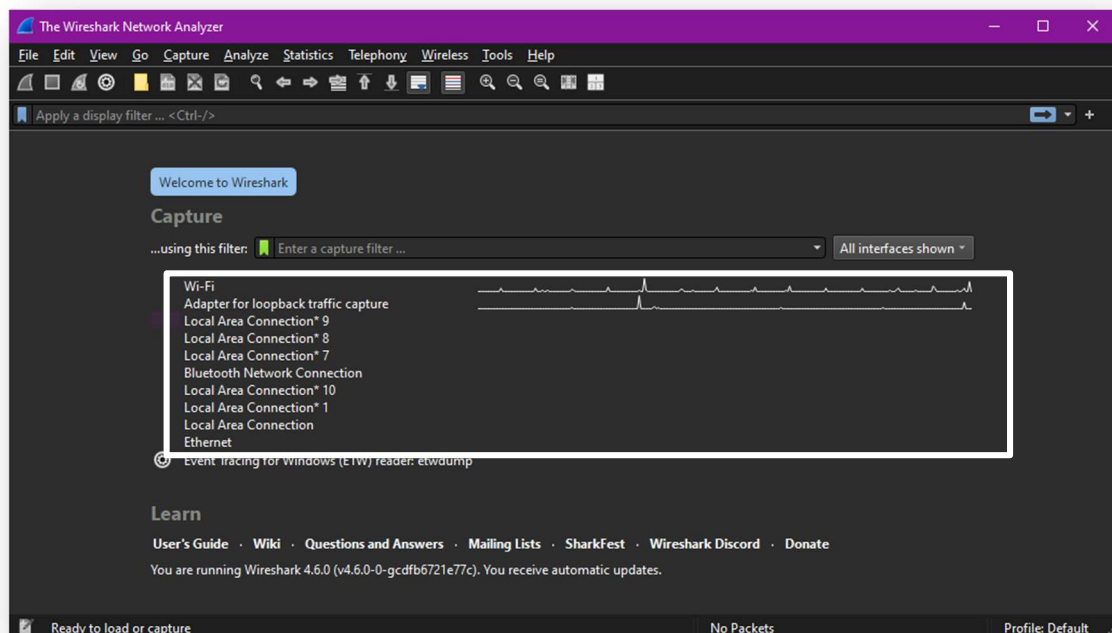


Figure 1. Wireshark home screen showing network interfaces

- Wireshark displays packets in real time and provides deep decoding of protocols such as **Ethernet**, **IP**, **TCP**, **UDP**, **ICMP**, **HTTP**, **DNS**, and many more.

❖ Steps to Capture Packets

1. **Start Wireshark** : Open Wireshark. It will display all **available network interfaces** (like Wi-Fi or Ethernet).

2. **Select an Interface :** Choose the network interface you want to capture traffic from (e.g., **Wi-Fi**). [Figure 1]
- Click **Start Capturing Packets**.

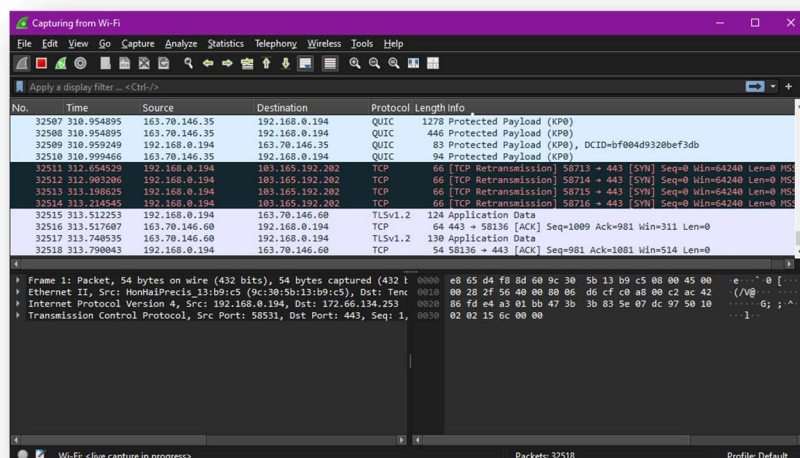
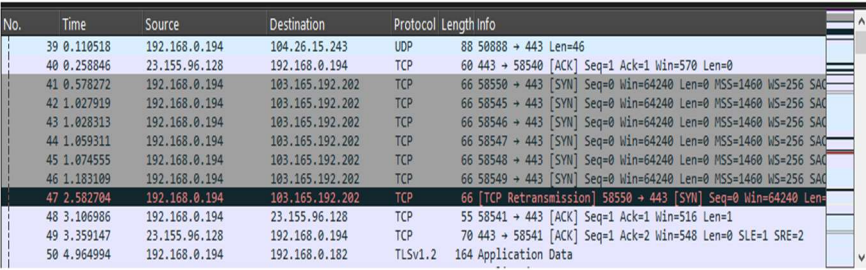
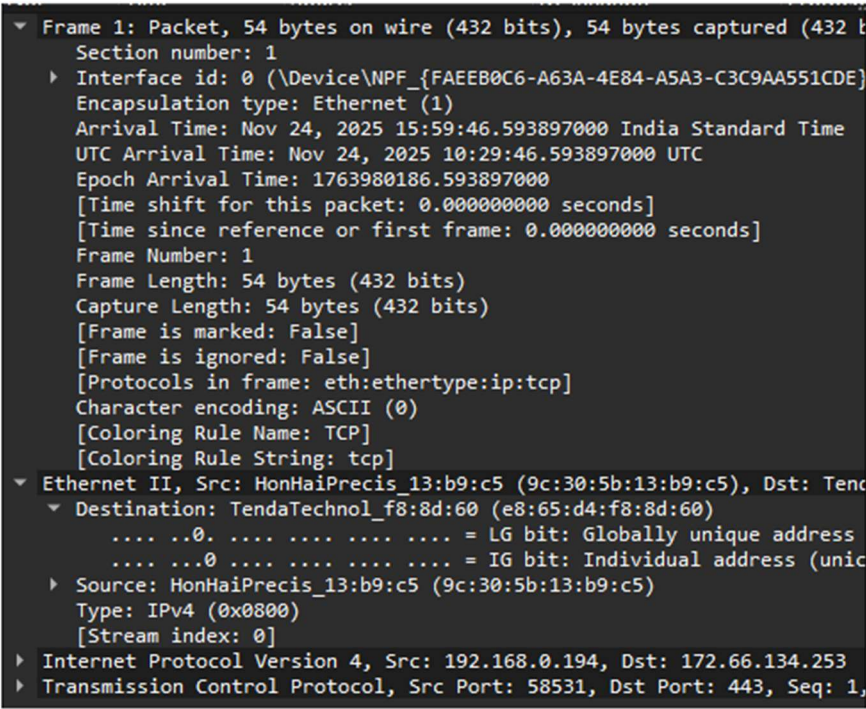
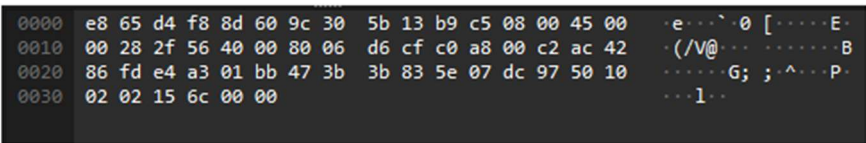


Figure 2. Live Packet Capture in Progress

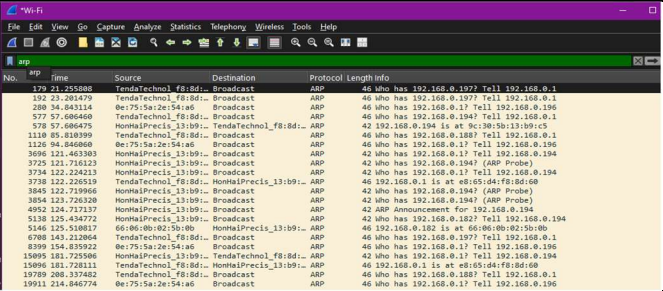
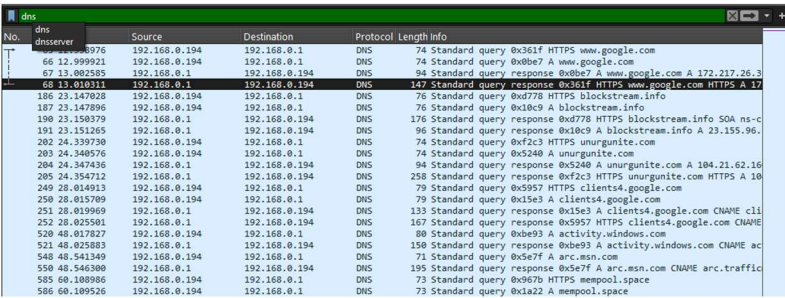
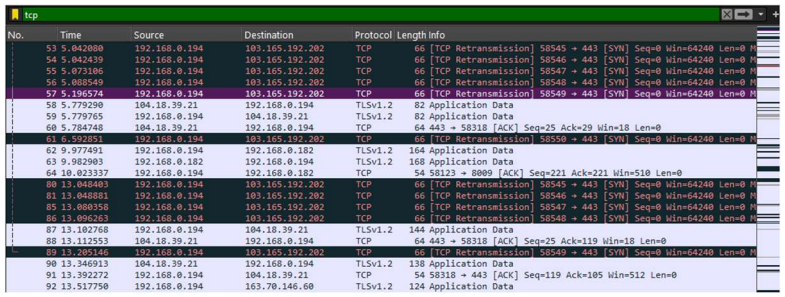
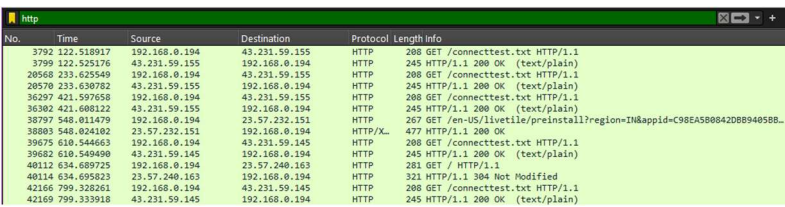
3. **Generate Network Traffic :** To capture meaningful traffic, open a few websites or run applications that use the internet.
- Examples:
- Open Google.com
 - Ping a website
 - Stream a video

Packets will start appearing in Wireshark. [Figure 2]

4. **Stop the Capture :** After collecting enough packets, click the Red Square (Stop Capture) button.
5. **Packet Analysis :** Wireshark displays packets in **three main panes**:

a) <u>Packet List Pane</u> :	Shows each captured packet with: • Packet number • Timestamp • Source & Destination IP • Protocol (TCP, UDP, DNS, HTTP, etc.) • Brief info	
b) <u>Packet Detail Pane</u> :	Shows detailed breakdown of layers: • Ethernet Frame • IP Header • TCP/UDP Header • Application Layer Protocol (DNS, HTTP, etc.)	
c) <u>Packet Bytes Pane</u> :	Displays raw packet data in hexadecimal and ASCII format.	

6. Analysis of Packets :

i. ARP Packet	- Used for mapping IP address to MAC address. - Shows "Who has X.X.X.X? Tell Y.Y.Y.Y"	
ii. DNS Packet	- Used to query domain names. - Example: Query for "www.google.com" Key fields: - Transaction ID - Query Type (A record) - Flags (standard query)	
iii. TCP Packet	Observations: - Source & destination ports - Sequence and acknowledgment numbers - Flags (SYN, ACK, FIN, etc.) Example: - SYN → Connection start - SYN-ACK → Response - ACK → Handshake completion	
iv. HTTP Packet	Shows communication between browser and server. Example: - HTTP GET request - Hostname - User-Agent - Response code (200 OK)	
v. ICMP Packet	Used for Ping. Types: - Echo Request - Echo Reply	